

AI in Finance and Risk Management

Credit risk

Complaint routing

Fair lending

Governance

Agenda

I	Where AI runs in a bank — five families of decisions	12 min
II	Case 1: the credit-risk review decision	15 min
III	Notebook: the credit model in five stages	20 min
IV	Explainability and fair lending	14 min
V	Case 2: complaint routing and the transformer	18 min
VI	Applied AI Studio: workflows and demos	14 min
VII	Governance and case studies	12 min
VIII	Wrap and the case-teardown assignment	5 min

SECTION I | 12 min

A bank runs on decisions, and every one has a price

Fraud, credit, money-laundering checks, customer service, trading. Five families of work — each with its own error cost and its own regulator.

Five families of decisions run a bank

FAMILY	THE DECISION BEING MADE	TONIGHT
Fraud detection	Is the transaction genuine? Millions of decisions at high speed.	Session 2 lab
Credit lifecycle	Who gets credit, on what terms, and which accounts need review?	Case 1 · today
Anti-money laundering	Which activity looks like hidden criminal money and must be reported to the government?	—
Customer service	Which team handles the request before its deadline?	Case 2 · today
Trading and markets	What should be bought, sold, or hedged?	—

Fraud and money-laundering checks share one lesson

Fraud: the Session 2 lesson

Score card transactions. A lower threshold catches more fraud but interrupts more honest customers. A higher threshold reverses the tradeoff. The confusion matrix makes both errors visible.

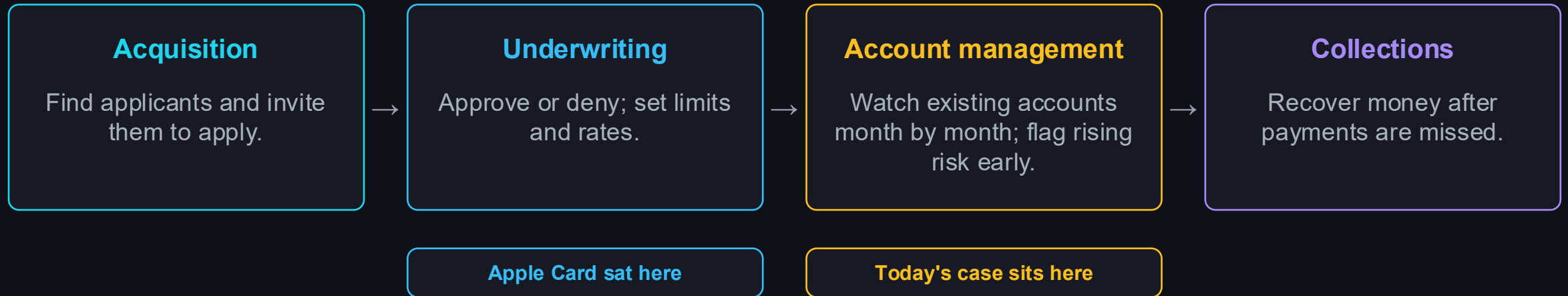
Anti-money laundering (AML)

Flag suspicious accounts and transfers for investigators. Most alerts prove innocent, but reviewing them still costs staff time. People review alerts before reporting.

The threshold sets the review workload

Missed cases and false alarms remain a tradeoff. Someone must choose the threshold and staff the queue.

Today's case sits inside the credit lifecycle



Same customer, four different decisions

Each stage has different data, models, and rules. Approving an applicant and reviewing an existing account are different problems.

Why finance adopted AI early, but deploys carefully

1 The data is already there

Every swipe, payment, and balance has been recorded for decades — clean, structured, and abundant.

2 The outcomes label themselves

A payment is either made or missed. The label arrives automatically next month, no annotator needed.

3 Every error has a price in dollars

A missed default costs the balance; a wasted review costs analyst time. You can compute what a model is worth.

4 And the densest rulebook in this course

Fair lending, adverse-action notices, and model-risk guidance make deployment slower than training.

SECTION II | 15 min

Case one: a credit-risk decision, end to end

Not who gets a credit card. Which existing accounts the risk team should look at this month — and what the bank would have to tell a customer afterward.

The decision: who gets reviewed this month

The problem

Analyst capacity is limited. Which existing accounts need attention before the next missed payment?

What the model outputs

One estimated probability of missing next month's payment. Rank accounts for attention, not people for judgment.

What actually happens

An analyst reviews the account and reasons. The next step may be outreach, a payment plan, a limit review, or no action.

What this system is not

No automatic approval, denial, or limit change. A score requests human attention; it is not a decision or an outcome.

30,000 real accounts, six months of behavior

One customer per row: September 2005 credit limit, demographics, and six months of bills, payments, and delays.

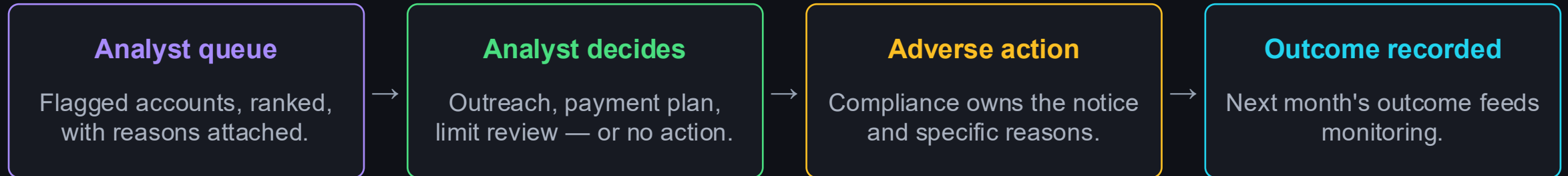
WHAT IS IN THE FILE	THE COLUMNS	WHY IT MATTERS TONIGHT
Exposure	Credit limit, in New Taiwan dollars	Stands in for the money at risk
Demographics	Sex, marriage, age, education	The fair-lending problem — slide 12
Payment status	How many months behind, for each of six months	The strongest signal, and easy to say out loud
Bills	Statement balance, each of six months	How much of the limit is being used
Payments	Amount actually paid, each of six months	How hard the customer is trying to keep up

From monthly snapshot to human decision

THE AUTOMATED HALF



THE HUMAN HALF



The authority boundary runs between the two rows

The model scores and flags. People choose the response. The model never contacts a customer, changes a limit, or declares a default.

The catch is sitting in the columns

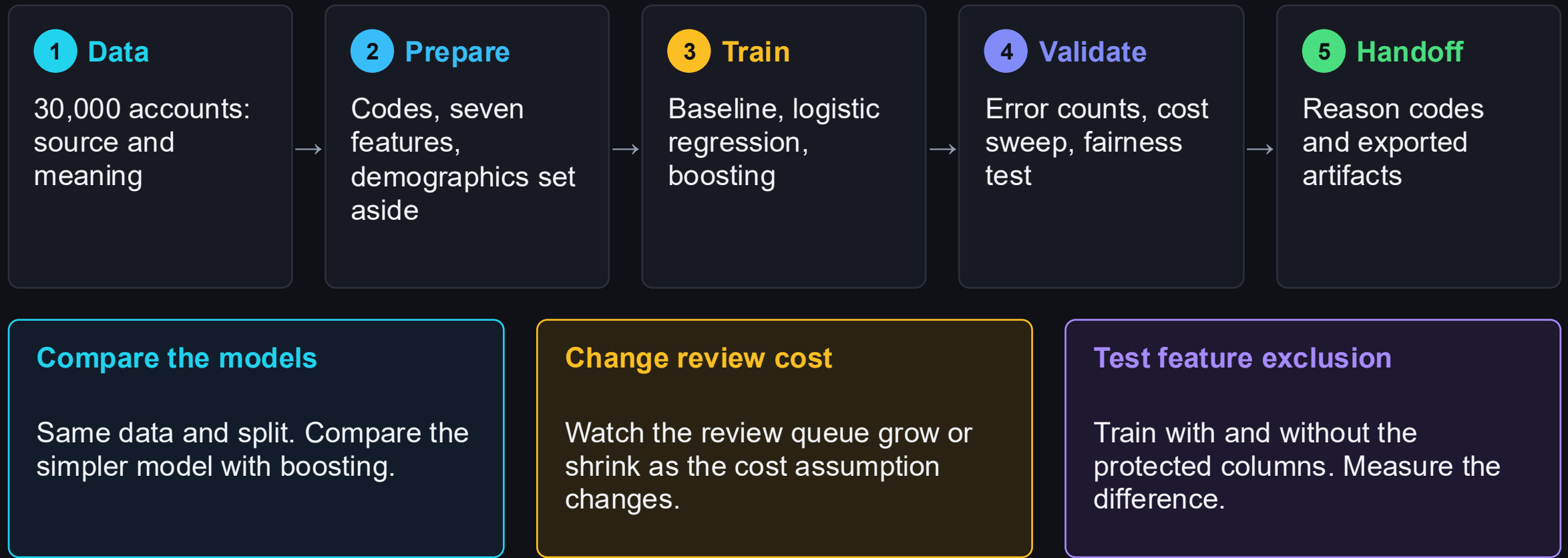
The dataset hands us sex, marriage, and age. A model would happily use them. US credit law says a credit decision may not.

COLUMN IN THE FILE	WHAT IT TELLS YOU	MAY A US CREDIT DECISION USE IT?
SEX	Male or female	Restricted by law (ECOA)
MARRIAGE	Married, single, other	Restricted by law (ECOA)
AGE	Age in years	Restricted by law (ECOA)
EDUCATION	Graduate school, university, high school	Not named, still risky
PAY_0 ... PAY_6	How many months behind, month by month	Yes — this is behavior
BILL_AMT, PAY_AMT	Billed and paid amounts, month by month	Yes — this is behavior

Open the credit notebook

OUR NOTEBOOK · From six months of card behavior to a review queue

notebooks/credit-risk/01_credit_build.ipynb



SECTION IV | 14 min

The model must explain itself. That's the law.

US credit law does not only ask whether the model is accurate. It asks whether the bank can tell one customer, in specific words, why the answer went against them.

Adverse action requires specific reasons

Denied credit, a reduced limit, or changed terms: the customer needs an explanation of the adverse decision.

1 The notice must give specific principal reasons

Regulation B, section 1002.9: state the main factors for this person, not a generic policy summary.

2 “Failed to achieve a qualifying score” is not enough

A low score alone is not a specific reason. Explain the factors behind it.

3 A complicated model is not a defense

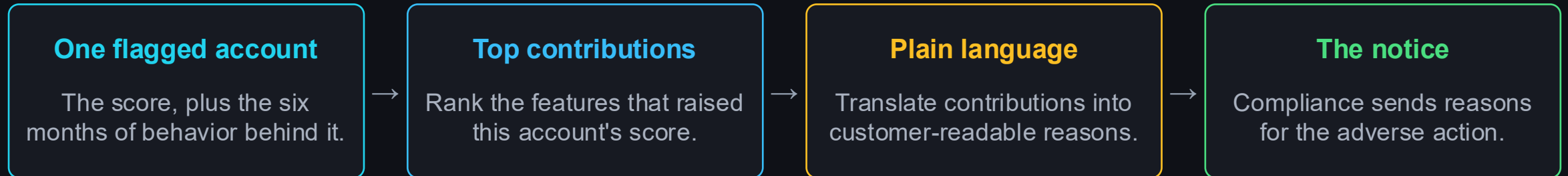
CFPB Circular 2022-03: not understanding the model does not excuse missing reasons. Complexity is not a defense.

Explainability is an engineering requirement

The system must answer why this account, not merely explain how the model works in general.

Sources: Regulation B, 12 CFR 1002.9 (adverse-action notifications) · CFPB Circular 2022-03

From a score to a sentence a customer can read



EXAMPLE · WHAT THE CUSTOMER WOULD ACTUALLY READ

“The principal reasons for this decision were: you were up to three months behind on payments in the last six months; you are currently two months behind; and you paid late in six of the last six months.”

The customer can check these statements against payment history. Example wording; the notebook generates the account-specific text.

Reason codes are why we built the features the way we did

Seven readable features can become sentences. A raw column name such as PAY_3 is not a useful customer explanation.

Remove the columns. Compare the scores.

Our classroom model leaves out sex, marital status, and age. We retrain without those columns and compare the results on the same accounts.

With sex, marriage, and age

AUC 0.7670

10 inputs. Same 6,000 validation accounts.

Without those three columns

AUC 0.7655

7 inputs. Same 6,000 validation accounts.

The score changed very little in this run

AUC fell by 0.0015. We keep the version with seven inputs and no demographic columns.

SECTION V | 18 min

Case two: same industry, but the data is words

A consumer writes a paragraph about what went wrong. Eight specialist teams could handle it, a regulatory clock is already running, and somebody has to decide where it goes.

Route each complaint to the right team

A REAL COMPLAINT, AS PUBLISHED BY THE REGULATOR

“According to my research, the title agency is supposed to provide my settlement agreement 3 days prior to close so I have time to review and question any details. It is 24 hours to close and I still have not received it.”

The narrative

The consumer's own words.



The model

Which of eight specialist teams does this belong to?



Confidence

High: auto-route. Low: human triage.



A team queue

Highlighted words explain the route.

The clock is already running, and misrouting spends it

Three days on the wrong desk consumes three days of the response deadline. Misrouting delays the consumer's answer.

The model does not judge whether the complaint is valid, does not draft the reply, and never contacts the consumer.

Narrative quoted from the CFPB Consumer Complaint Database — public US government data; narratives are opt-in and scrubbed of personal details before publication.

Real text, with real mess in it

1 Narratives are opt-in

Our 2023+ window held 2,634,602 narratives. These consumers chose publication; they do not represent everyone.

2 One team swamps all the others

Credit reporting: 2,664 of 8,728 test complaints. Always choosing it gives 30.5% accuracy but only 0.0585 macro-F1.

3 41.5% of the recent narratives are exact duplicates

1,093,131 copied rows out of 2,634,602. Remove duplicate letters before splitting, or testing rewards memorization.

4 The category names changed over the years

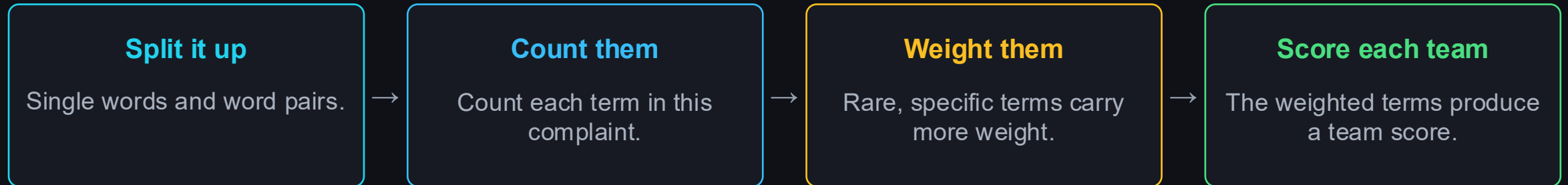
Map renamed product labels onto eight teams, and document the mapping.

Data preparation sets the model's limits

Document each choice. Macro-F1 balances precision and recall within each team, then averages the eight team scores equally.

Way one: count the words

First turn the paragraph into numbers. Count words and word pairs, then give more weight to distinctive terms.



WHY THE COMPLAINT ABOVE WENT TO THE MORTGAGE TEAM

mortgage · loan · foreclosure · modification · equity

The highest-weighted words in that complaint for that team. The explanation is not a second system bolted on afterward — the explanation is the model.

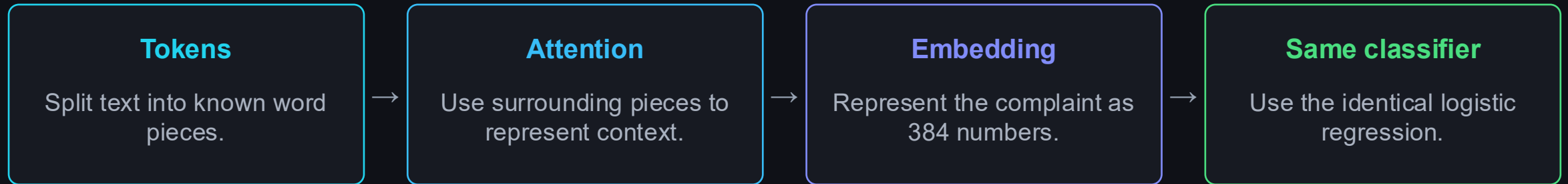
This is the model the bank actually deploys

Fast laptop training, offline inference, and visible routing words. The transformer is a comparison, not a replacement.

Method: TF-IDF word and word-pair features, then logistic regression — the same classifier used in the transformer comparison.

Way two: the transformer

Counting words treats “the bank charged me a late fee” and “a late fee charged me the bank” as identical. A transformer reads each word in the company of every other word.



all-MiniLM-L6-v2 | SETTINGS READ FROM THE MODEL

6 transformer layers · 12 attention heads in each layer · hidden size 384 · 22,713,216 parameters · reads the first 256 word pieces, then stops · 183.2 MB to download

Runs on a classroom CPU. Embeddings are precomputed for the offline lab.

An embedding is a better ruler, not a better judge

Similar meanings get similar representations. That does not establish legal judgment, complaint validity, or word-level reasons.

Model: sentence-transformers/all-MiniLM-L6-v2, downloaded once and cached; embeddings are precomputed so the lab needs no network.

Matched comparison: explainability breaks the tie

15,000 training complaints; 8,728 validation complaints; identical classifier. Only the text representation changes.

TF-IDF + logistic regression

ACCURACY ▲ AHEAD

80.83%

MACRO-F1

0.7918

7.6 s fit. Visible word-level contributions.

MiniLM + the same classifier

ACCURACY

80.48%

MACRO-F1 ▲ AHEAD

0.7957

1.1 s fit, excluding encoding and the 183.2 MB download.

A representation trade, not an upgrade

Word counts lead accuracy by 0.35 percentage points; embeddings lead macro-F1 by 0.0039. Visible reasons and offline simplicity decide the choice.

SECTION VII | 12 min

Who lets a model into production?

Training was the easy part. Now somebody has to sign their name to the claim that this model may be used — and keep watching it after it ships.

A model is a risk, and risks get managed

US model-risk guidance connects directly to the artifacts and workflow we built tonight.

- 1 Write it down**
Intended use, data, performance, and limitations. Start with the model card.
- 2 Somebody else checks it**
Independent validators challenge the builder's work and can say no.
- 3 Watch it after it ships**
Track changes in customers, products, the economy, and model performance.
- 4 Keep an inventory**
List each model, its owner, version, and review date.

The guidance itself gets maintained like a model

SR 11-7 (2011) was superseded on April 17, 2026 by SR 26-2 — governance is not a document you write once and file.

Apple Card: no violation found, real damage done

In November 2019 a viral thread accused the Apple Card of giving a husband twenty times the credit limit of his wife. New York's financial regulator opened an investigation.

The allegation

Shared finances, very different limits, and no clear explanation for the customers.

What the regulator found

About 400,000 New York applicants reviewed. No evidence of deliberate or disparate-impact discrimination found.

What it criticized anyway

Poor customer service and transparency. No adequate process to explain individual limits.

You can be lawful and still fail the customer

A defensible model still needs a usable explanation and a process for answering customer questions.

CLICK TO OPEN PRIMARY SOURCES

[DFS press release ↗](#)

[DFS full investigation report ↗](#)

Source: New York State Department of Financial Services, report on the Apple Card investigation (March 2021)

One failure, one success with an asterisk

Knight Capital · 2012

A faulty trading-software deployment caused more than \$460 million in losses in about 45 minutes; the SEC imposed a \$12 million penalty. No machine learning was involved. Release checks and stop controls still matter.

Upstart · 2017–2022

CFPB issued a no-action letter for AI underwriting, with fair-lending reporting. Upstart requested termination in June 2022 because model changes required review under the arrangement.

The model is never the whole system

Knight: control deployment. Upstart: oversight can constrain model updates. Neither lesson is about accuracy alone.

CLICK TO OPEN PRIMARY SOURCES

[SEC press release · Knight ↗](#)

[CFPB · first no-action letter ↗](#)

[CFPB · termination order ↗](#)

Sources: SEC press release and administrative order (2013) · CFPB newsroom (2017 and 2022)

Five things to carry out of tonight

- 1 Name the decision**
Place the task in fraud, credit, AML, service, or trading. Name its error cost and oversight.
- 2 The model ranks; a person decides**
Keep the score, the decision, and the eventual outcome separate.
- 3 Explanation is a legal artifact**
Specific reasons must be readable by the customer. Complexity is not an excuse.
- 4 Newest is not automatically best**
Match the comparison. Consider performance, explainability, and cost together.
- 5 Governance decides what actually ships**
Document, validate independently, monitor, and inventory the model.